Signal Attenuation Through a Lossy Cable: Fitting the Decay Model

D. Measure

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Abstract

A tone is injected into a lossy cable and the received amplitude is recorded against distance. We fit the measured data to the two-parameter decay model and report the residual of the fit.

1 The model

The received amplitude follows the exponential decay law

$$y = a\,e^{-bx}, \tag{1}$$

where x is distance in metres, y the measured amplitude, a the launch amplitude and b the attenuation constant.

2 Measurements

The bench data used for the fit are:

x (m)	y (mV)
0.0	1.98
1.0	1.21
2.0	0.74
3.0	0.45
4.0	0.28

3 Fit quality

The least-squares fit of the model to the measured data above is assessed by its residual pattern; a systematic residual indicates a non-exponential loss channel.